FOUNDATION COURSE

START YOUR

PREPARATION

FROM BASICS

Prerequisites of Engineering Mathematics

Lecture no. 03

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Recap of previous lecture

(Inequalities & Mod function)



Topics to be covered

"LOGARITHMS"



Concept: eg $\log_2 8 = ? = 3 \iff (2^3 = 8)$

eg $\log_{\sqrt{2}}(8) = ? = 6 \quad (\because (\sqrt{2})^6 = 8)$

eg $\log_7(1) = ? = 0 \quad (\because 7^0 = 1)$

eg $\log_5(625) = ? = 4 \quad (\because 5^4 = 625)$

eg $\log_{10}(0.01) = ? = -2 \quad (\because 10^{-2} = \frac{1}{100} = 0.01)$

eg $\log_5(0.008) = ? = -3 \quad \left(\because 0.008 = \frac{8}{1000} = \frac{1}{125} = 5^{-3}\right)$

eg: $\log_{(2\sqrt{3})} 1728 = x$ then $x = ?$

$\Rightarrow (2\sqrt{3})^x = 1728 = (12)^3$

$= (4 \times 3)^3 = [(2\sqrt{3})^2]^3$

$(2\sqrt{3})^x = (2\sqrt{3})^6 \Rightarrow x = 6$

i.e $\log_{(2\sqrt{3})} 1728 = 6$

Def^n: if $\boxed{\log_a x = b \iff a^b = x}$ $\because \log_4(64) = 3 \Leftrightarrow 4^3 = 64$

Here a is taken as the number i.e. $(a>0)$, $\Rightarrow a^b > 0$ i.e. $(x>0)$ But b may be $-ve$.

Note:- ① $\boxed{\log_a a = 1}$, $a>0$ & $a \neq 1$

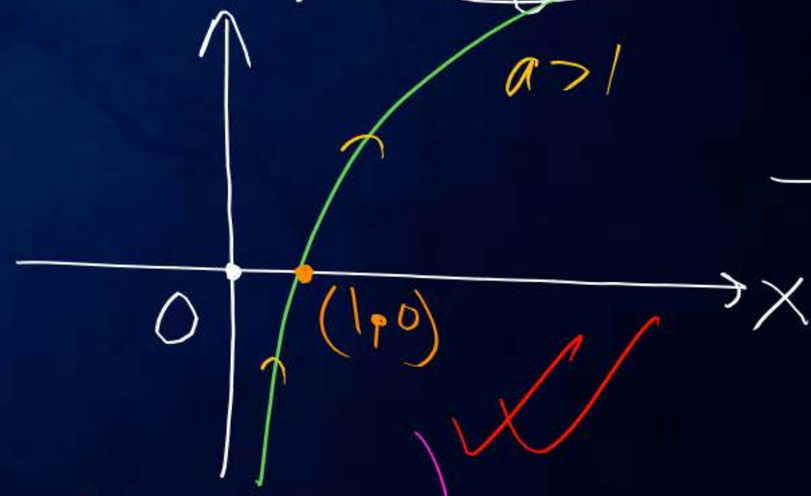
eg $\log_7 7 = 1$, $\log_5 5 = 1$, $\log_1 1 = N.D$

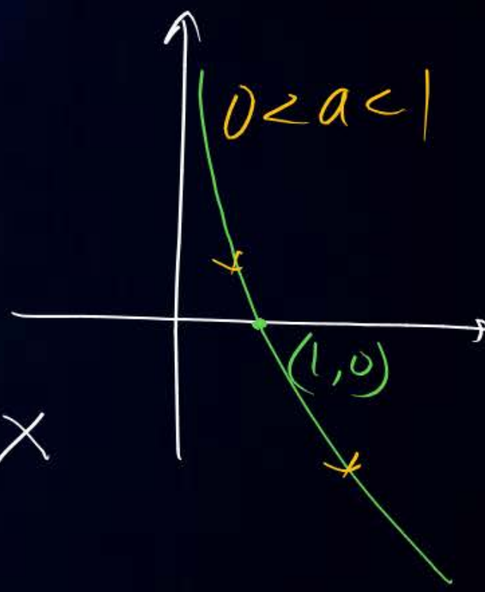
② $\boxed{\log_a 1 = 0}$,

③ $\boxed{\log_a(\infty) = \infty}$ (for $a>1$)

④ $\boxed{\log_a(0) = -\infty}$ (for $a>1$) $\left(\because a^{-\infty} = \frac{1}{a^{\infty}} = \frac{1}{\infty} = 0\right)$

⑤ Graph of $(y = \log_a x)$.





⑥ $\log_a x = \log_a y \iff x = y$

Proof: Let $\log_a x = \alpha \Rightarrow a^\alpha = x$

& $\log_a y = \alpha \Rightarrow a^\alpha = y$

$\Rightarrow x = y$

⑦ if $a > 1$, $\log_a x > \log_a y \iff x > y$

if $0 < a < 1$, $\log_a x > \log_a y \iff x < y$

⑧ Product formula

$$\log_a(mn) = \log_a m + \log_a n$$

Let $\log_a m = \alpha \Rightarrow a^\alpha = m$

& $\log_a n = \beta \Rightarrow a^\beta = n$

Now $a^{\alpha+\beta} = a^\alpha \cdot a^\beta = m \cdot n$

$\because$ when $2^3 = 8 \Rightarrow \log_2 8 = 3$

$\log_a(mn) = \alpha + \beta = \log_a m + \log_a n$

(9) Quotient Formula –

$$\log_a\left(\frac{m}{n}\right) = \log_a(m) - \log_a(n)$$

Proof: Let $\log_a(m) = \alpha \Longrightarrow a^{\alpha} = m$

& $\log_a(n) = \beta \Longrightarrow a^{\beta} = n$

Now $a^{\alpha-\beta} = a^{\alpha} \cdot a^{-\beta} = \frac{a^{\alpha}}{a^{\beta}} = \frac{m}{n}$

$\because 2^3 = 8 \Longrightarrow \log_2 8 = 3$

$$\log_a\left(\frac{m}{n}\right) = \alpha - \beta = \log_a m - \log_a n$$

(10) Power formula –

$$\log_a(m^n) = n \cdot \log_a m$$

Proof: $\log_a(m^n) = \log_a(\underbrace{m \cdot m \cdot m \cdots m}_{n \text{ times}})$

$= \underbrace{\log_a m + \log_a m + \log_a m + \cdots + \log_a m}_{n \text{ times}}$

$= n \log_a m$

(11) Base Change formula →

$$\log_y(x) = \frac{\log_a x}{\log_a y}$$

Let $\log_y(x) = \alpha \implies y^\alpha = x$

Now $\log_a(x) = \log_a(y^\alpha) = \alpha \log_a(y)$

$$\log_a x = \log_y x \cdot (\log_a y)$$

i.e. $$\log_y x = \frac{\log_a x}{\log_a y} \quad //$$

(*) Why $\log_1 x = \text{N.D}$?

Reason: $$\log_1 x = \frac{\log_a x}{\log_a 1} = \frac{\log_a x}{0} = \text{ND}$$

(*) Why $\log_1 1 = \text{ND}$?

Reason: $$\log_1 1 = \frac{\log_a 1}{\log_a 1} = \frac{0}{0} = \text{IND form}$$

(12) Reciprocal formula →

$$\log_y x = \frac{\log_x x}{\log_x y} = \frac{1}{\log_x y}$$

$$\log_y x = \frac{1}{\log_x y}$$

(13) $\log_{a^b}(x) = \frac{1}{b}\log_a x$

LHS $= \log_{a^b}(x) = \frac{\log_a(x)}{\log_a(a^b)} = \frac{\log_a x}{b \log_a a}$

$= \frac{1}{b}\log_a x = RHS$

(14) $a = e^{\log_e a}$ if $a > 0$

Let $e^{\log_e a} = \alpha \Longrightarrow \log_e \alpha = \log_e a \Rightarrow \alpha = a$

$\therefore \left(2^3 = 8 \Longrightarrow \log_2 8 = 3\right)$

i.e. $e^{\log_e a} = a$

(15) $a^x = e^{x \log_e a}$; $a > 0$

W.K.T, $a = e^{\log_e a}$

$\Rightarrow a^x = e^{\log(a^x)} = e^{x \log_e a}$ Hence Proved

(16) $a^{\log_e b} = b^{\log_e a}$

$a^{\log_e b} = \left(e^{\log_e a}\right)^{\log_e b} = e^{\log a \cdot \log b} = e^{\log_e b \cdot \log_e a}$

$= \left(e^{\log_e b}\right)^{\log_e a} = b^{\log_e a}$

$=$

Q: $\log_{10} x - \log_{10}(2x-1) = 1$

then solve for $x = ?$

Sol:

$$\log_{10}\left(\frac{x}{2x-1}\right) = 1$$

$$\frac{x}{2x-1} = 10^1$$

$$x = 20x - 10$$

$$19x = 10 \Rightarrow x = \frac{10}{19}$$

Q: $2\log_3\left[1+\log_2\left(1+3\log_3 x\right)\right] = 2$

W.K. that $\log_a x = y \Rightarrow x = a^y$

$$\log_3\left[1+\log_2\left(1+3\log_3 x\right)\right] = 1$$

$$1+\log_2\left(1+3\log_3 x\right) = 3^1$$

$$\log_2\left(1+3\log_3 x\right) = 2$$

$$1+3\log_3 x = 2^2$$

$$3\log_3 x = 3$$

$$\log_3 x = 1 \Rightarrow x = 3^1 = 3$$

Q. if $a=\log_x(yz)$, $b=\log_y(zx)$ & $c=\log_z(xy)$ then $\frac{1}{1+a}+\frac{1}{1+b}+\frac{1}{1+c}=?$

$$\frac{1}{1+a}=\frac{1}{1+\log_x(yz)}=\frac{1}{\log_x x+\log_x(yz)}=\frac{1}{\log_x(xyz)}=\log_{xyz}(x)$$

Similarly $\frac{1}{1+b}=\frac{1}{1+\log_y(zx)}=\frac{1}{\log_y y+\log_y(zx)}=\frac{1}{\log_y(xyz)}=\log_{xyz}(y)$

& $\frac{1}{1+c}=\frac{1}{1+\log_z(xy)}=\frac{1}{\log_z z+\log_z(xy)}=\frac{1}{\log_z(xyz)}=\log_{xyz}(z)$

So $\frac{1}{1+a}+\frac{1}{1+b}+\frac{1}{1+c}=\log_{xyz}(x)+\log_{xyz}(y)+\log_{xyz}(z)=\log_{xyz}(xyz)=1$


Thank
YOU